

Multiple Linear Regression

“The height of sophistication is simplicity.”
— Clare Boothe Luce

 Statistical Reasoning Lecture #2
Alexander Savi, 2025

 Mehmetoglu & Mittner Ch. (4, 5), 8, 10.1, .2, .4, .7, .8

Untitled by Koen Derks (aRtsy package) 



News

INTERVIEW

“Trump neemt risico’s op basis van gevoel, niet op basis van analyse. Het is allemaal improvisatie”

Nate Silver | statisticus, pokerspeler Nate Silver kijkt als cijferaar naar de Amerikaanse politiek en samenleving. Hij gelooft in risico’s nemen om te willen winnen. „De rijkste mensen van de wereld verdubbelen hun rijkdom elke tien jaar.”



Bas Blokker

1 september 2025 om 15:48 • Leestijd 7 minuten

— [NRC](#) (Sep. 1, 2025)



Contact

How

Direct

- ☐ Lecture (WF/SR)
- ☐ Tutorial (WF)
- ☐ Consultation hour (SR)

Indirect (if not otherwise possible)

- ☐ Canvas or email

Who

Is it about WF?

- ☐ Tutorial teacher (initial point of contact)
- ☐ Jolien (substantive questions, during lecture)

Is it about SR?

- ☐ Substantive: Alexander
- ☐ Weekly assignments: Jonas

Is it cross-course?

- ☐ Alexander (course coordinator)

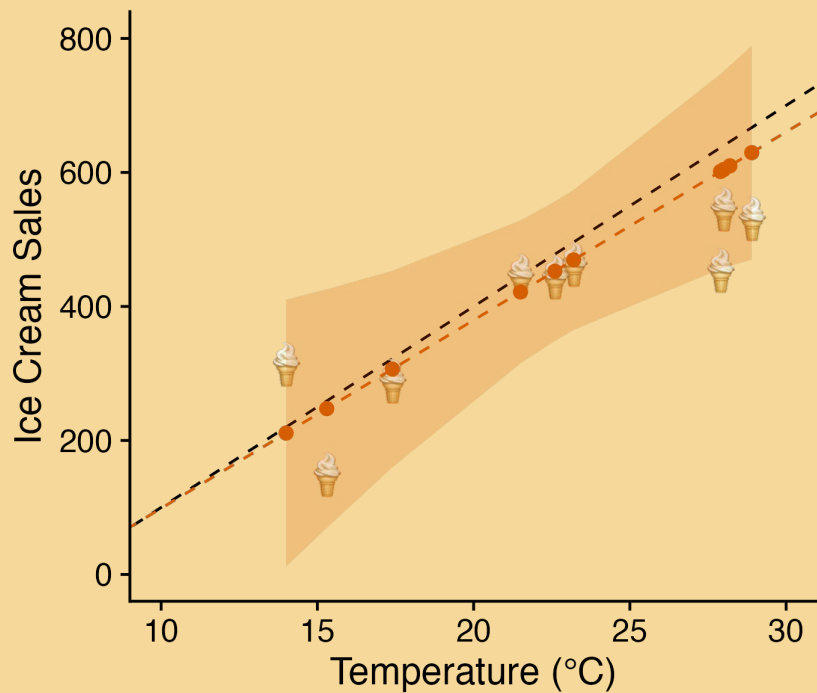


Announcements

- ❑ [SMASH](#)
- ❑ Web lectures & attendance
- ❑ Personal course manual (for learning, can't bring to exam)
- ❑ [Online consultation](#) with Jonas (restricted access to UvA accounts)



Recap



- RMSE
- Explained variance R^2
- t statistic
- Effect size f^2
- Confidence interval



Today

Topics

Statistical reasoning with GLM

Multiple linear regression

| Multiple linear regression

| Moderation/interaction analysis

Dummy-variable regression

Logistic regression

Multilevel and longitudinal analysis

Bayesian statistics

Learning goals

Estimate the relationships between more than two continuous variables.

Determine whether the relationship between two continuous variables depends on a third continuous variable.

Multiple Linear Regression

Iris



Sepal
Kelkblad

Petal
Kroonblad

Q. Are the dimensions of the petals and sepals of the iris flower related?

H. The length of a petal is related to the length and the width of a sepal.

E. [...]



A data set made famous by [Ronald Fisher](#) and with its very own [Wikipedia page](#).

More than one (continuous) independent variables



Willem the Supervillain

I'm on it, let's move fast and break things.



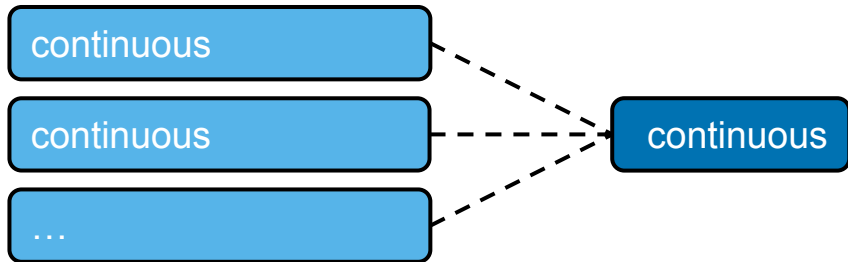
Miró the Superhero

Sure, let's move carefully and fix things.



*I have never tried that before, so I think I should
definitely be able to do that.*

Model thinking



Let's just add them and see what happens.



$$\text{Petal.Length} = \beta_0 + \beta_1(\text{Sepal.Length}) + \beta_2(\text{Sepal.Width}) + \epsilon$$

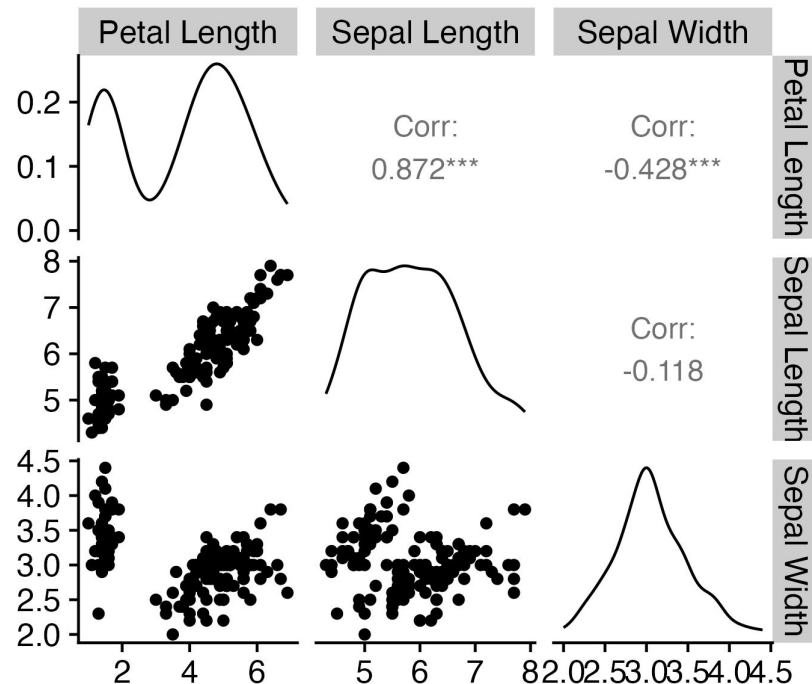
```
mod <- Petal.Length ~ Sepal.Length +  
Sepal.Width
```

		Independent	
		Categorical	Continuous
Dependent	Cate gori cal		
	Con tinu ous		Simple regression Multiple regression

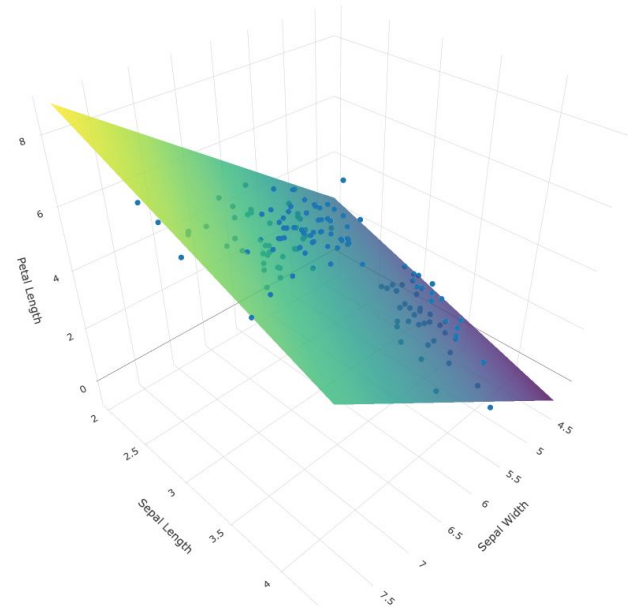
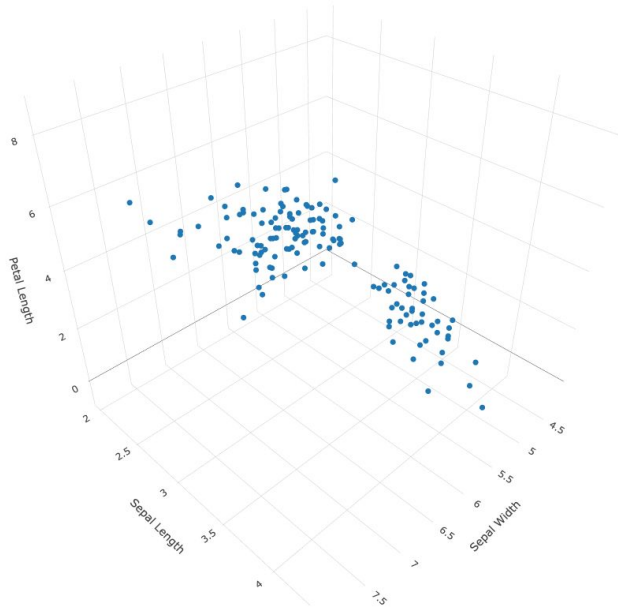
Data & transformations

```
data(iris)
help(iris)
head(iris)
str(iris)
pairs(iris)
```

```
'data.frame': 150 obs. of 5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num   3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num   1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num   0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species      : Factor w/ 3 levels "setosa","versicolor",...: 1 1 1 1 1 1 1 1 1 1 ...
```



Data & transformations



Results

```
fit <- lm(formula = mod, data = iris)
summary(fit); resid(fit); confint(fit)
```

Call:

```
lm(formula = mod, data = iris)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.25582	-0.46922	-0.05741	0.45530	1.75599

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.52476	0.56344	-4.481	1.48e-05 ***
Sepal.Length	1.77559	0.06441	27.569	< 2e-16 ***
Sepal.Width	-1.33862	0.12236	-10.940	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6465 on 147 degrees of freedom

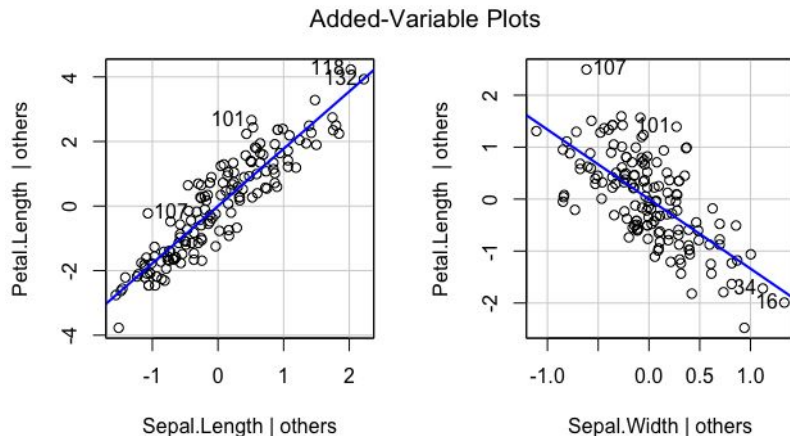
Multiple R-squared: 0.8677, Adjusted R-squared: 0.8659

F-statistic: 482 on 2 and 147 DF, p-value: < 2.2e-16

$$\widehat{\text{Petal.Length}} = -2.52 + 1.78(\text{Sepal.Length}) - 1.34(\text{Sepal.Width})$$

Partial slope coefficient

Hey, Willem the Supervillain, the meanings of your slope coefficients have changed!

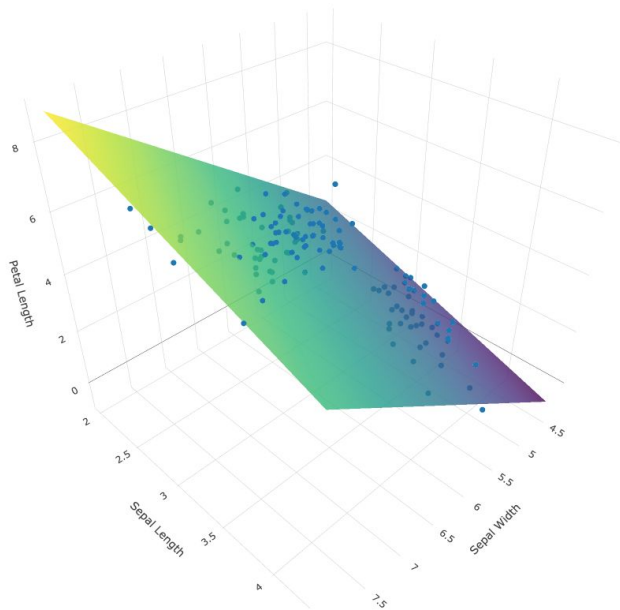


“The predicted mean Petal Length (Y) changes by 1.78 (predicted β_1) for every unit increase in Sepal Length (X_1), *having controlled for Sepal Width (X_2)*.”

“| others” = *holding the other variables constant*

```
library("car")
car::avPlots(fit)
```

Ms. Signal 🧑🏻🧑🏻 Mr. Noise

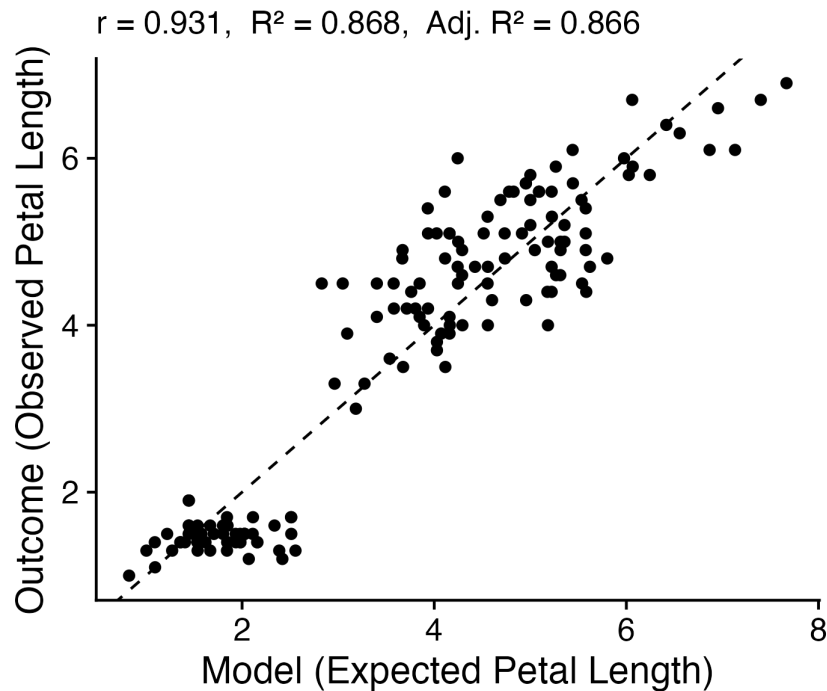


Evaluation | (Adjusted) R^2

Hey, Willem the Supervillain, be careful, your explained variance is inflated!



Penalty for increasing the number of independent variables, more so for smaller n .



```
observed <- iris$Petal_Length  
expected <- fitted(fit)  
cor(observed, expected)^2
```

Evaluation | Model comparison

Statistical significance

Explained variance (multiple R^2)

Predictive validity

Effect size

```
# model comparison
mod_0 <- Petal.Length ~ Sepal.Length
fit_0 <- lm(formula = mod_0, data = iris)
anova(fit, fit_0)
```

Standardization

- ❑ For comparison of slope coefficients (if expressed in different or unintuitive measurements)
- ❑ What about iris data set?

```
iris_std <- iris |>
  mutate(across(where(is.numeric), scale))
fit_std <- lm(formula = mod, data = iris_std)
summary(fit_std)
```

Unstandardized

The mean of petal length changes by 1.78 centimeters for every centimeter increase in sepal length, having controlled for sepal width.

$$\widehat{\text{Petal.Length}} = -2.52 + 1.78(\text{Sepal.Length}) - 1.34(\text{Sepal.Width})$$

Standardized

The mean of petal length changes by .83 standard deviations for every standard deviation increase in sepal length, having controlled for sepal width.

$$\widehat{\text{Petal.Length}} = 0 + 0.83(\text{Sepal.Length}) - 0.33(\text{Sepal.Width})$$



R is the shit

```
library("report")
report(fit, alpha = .001)
```

We fitted a linear model (estimated using OLS) to predict `Petal.Length` with `Sepal.Length` and `Sepal.Width` (formula: `Petal.Length ~ Sepal.Length + Sepal.Width`). The model explains a statistically significant and substantial proportion of variance ($R^2 = 0.87$, $F(2, 147) = 482.00$, $p < .001$, adj. $R^2 = 0.87$). The model's intercept, corresponding to `Sepal.Length = 0` and `Sepal.Width = 0`, is at -2.52 (95% CI $[-3.64, -1.41]$, $t(147) = -4.48$, $p < .001$). Within this model:

- The effect of `Sepal.Length` is statistically significant and positive ($\beta = 1.78$, 95% CI $[1.65, 1.90]$, $t(147) = 27.57$, $p < .001$; Std. $\beta = 0.83$, 95% CI $[0.77, 0.89]$)
- The effect of `Sepal.Width` is statistically significant and negative ($\beta = -1.34$, 95% CI $[-1.58, -1.10]$, $t(147) = -10.94$, $p < .001$; Std. $\beta = -0.33$, 95% CI $[-0.39, -0.27]$)

Standardized parameters were obtained by fitting the model on a standardized version of the dataset. 95% Confidence Intervals (CIs) and p-values were computed using a Wald t-distribution approximation.

Moderation / Interaction Analysis

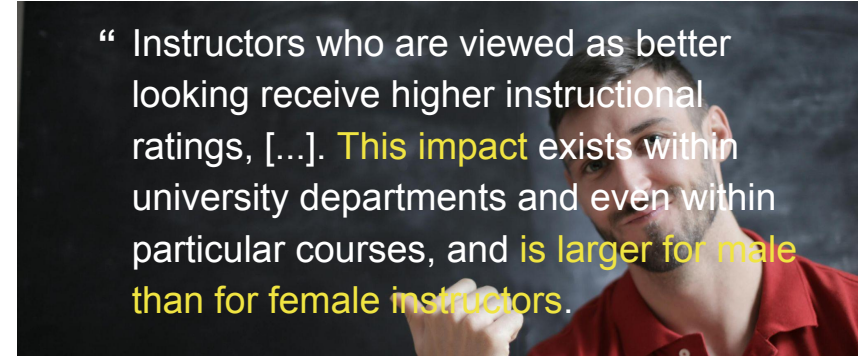
Student evaluations

De student als consument maakt vrouwelijke docenten extra kwetsbaar

Nieuws | door Frans van Heest

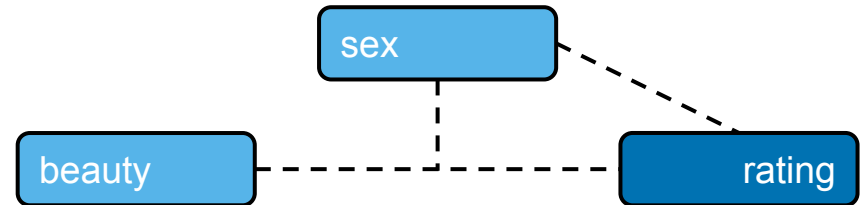
13 september 2023 | Vrouwelijke docenten worden aantoonbaar gediscrimineerd door studentenevaluaties, maar toch blijft het instrument voor veel universiteiten belangrijk om medewerkers te beoordelen. Cursusevaluaties moedigen echter middelmatig onderwijs aan en zijn extra nadelig voor vrouwen.

— [ScienceGuide](#)  (Sep. 13, 2023)



— [Hamermesh & Parker \(2005\)](#)  ; [NBER](#)

— Photo by [Andrea Piacquadio](#)



Main effects, interaction effects



Willem, did you know we've been analyzing main effects, using 'linear additive models'?

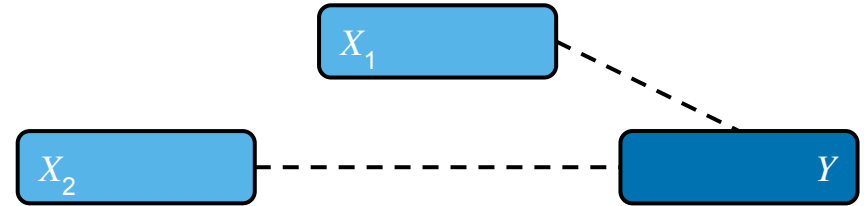


Sure, whatever, let's move on. I'm going to do interactions and start multiplying terms.

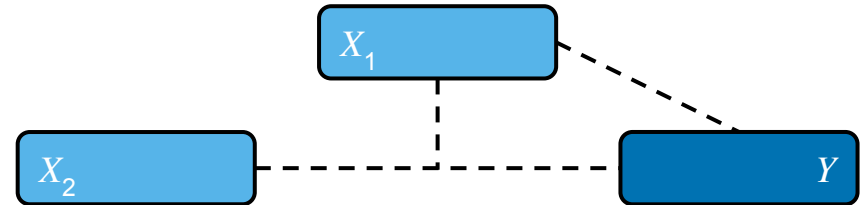


If this is what you call an interaction, we're definitely not going to multiply.

Additive



Nonadditive



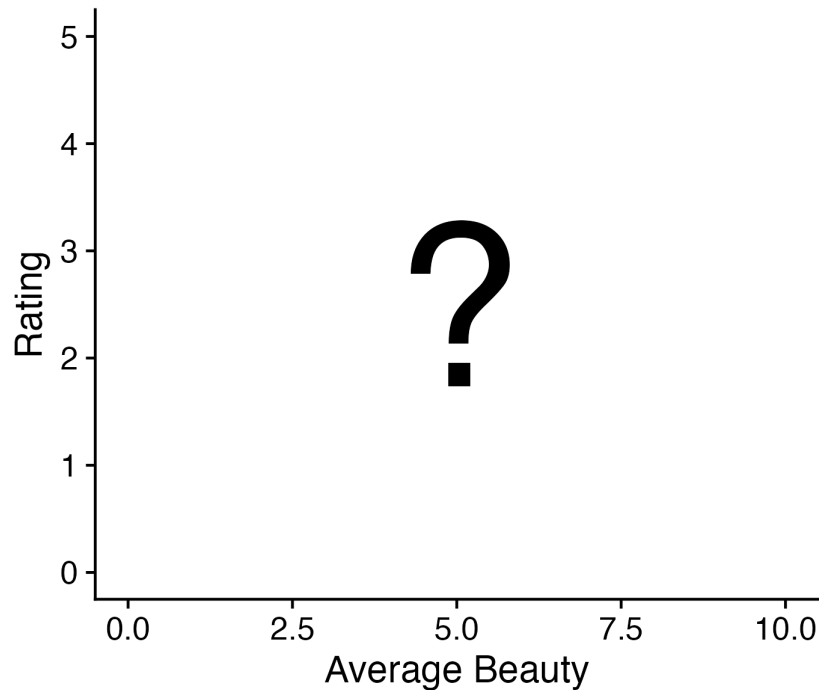
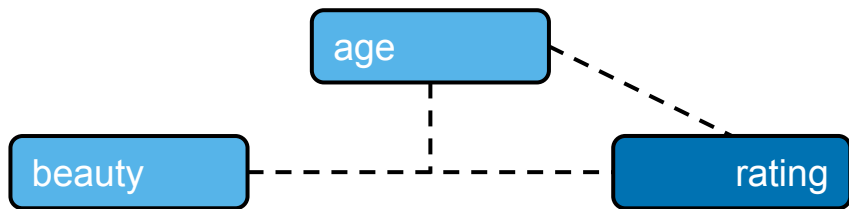


Interactions

Q. Is the effect of beauty on instructional rating modified by age?

H. The effect of beauty on rating is larger for [younger/older] teachers than for [younger/older] teachers.

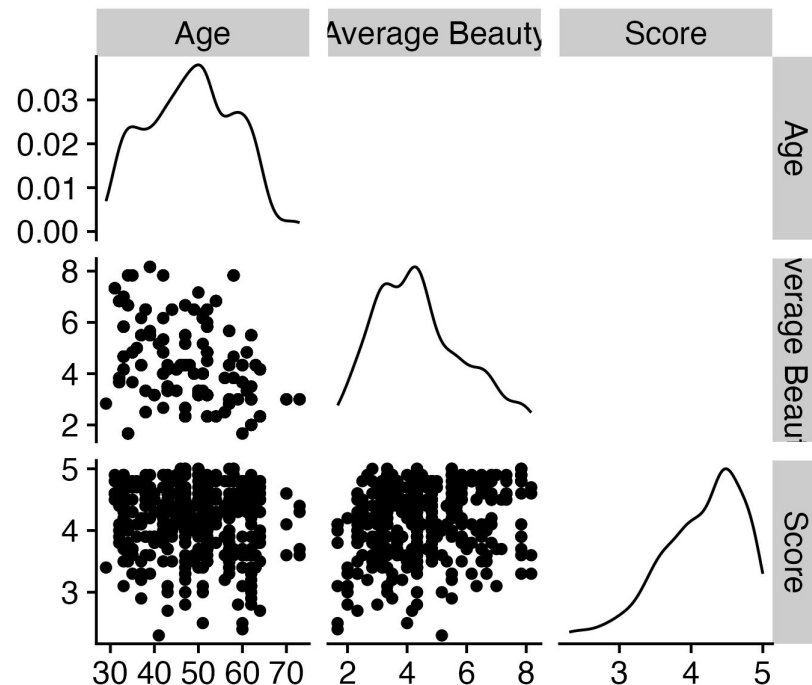
E. ...



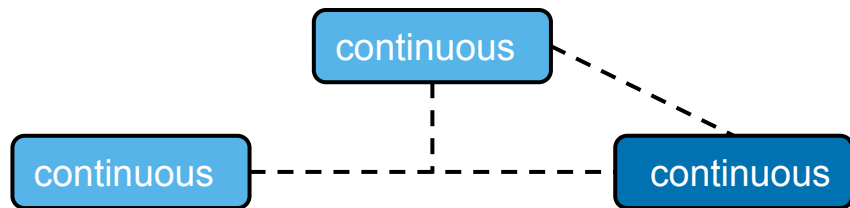
Data

```
library("moderndive")
data(package = "moderndive")
data(evals)
help(evals)
pairs(evals)
str(evals)
```

```
tibble [463 × 14] (S3: tbl_df/tbl/data.frame)
 $ ID          : int [1:463] 1 2 3 4 5 6 7 8 9 10 ...
 $ prof_ID     : int [1:463] 1 1 1 1 2 2 2 3 3 4 ...
 $ score       : num [1:463] 4.7 4.1 3.9 4.8 4.6 4.3 2.8 4.1 3.4 4.5 ...
 $ age        : int [1:463] 36 36 36 36 59 59 59 51 51 40 ...
 $ bty_avg     : num [1:463] 5 5 5 5 3 ...
 $ gender      : Factor w/ 2 levels "female","male": 1 1 1 1 2 2 2 2 2 1 ...
 $ ethnicity   : Factor w/ 2 levels "minority","not minority": 1 1 1 1 2 2 2 2 2 2 ...
 $ language    : Factor w/ 2 levels "english","non-english": 1 1 1 1 1 1 1 1 1 1 ...
 $ rank        : Factor w/ 3 levels "teaching","tenure track",...: 2 2 2 2 3 3 3 3 3 3 ...
 $ pic_outfit  : Factor w/ 2 levels "formal","not formal": 2 2 2 2 2 2 2 2 2 2 ...
 $ pic_color   : Factor w/ 2 levels "black&white",...: 2 2 2 2 2 2 2 2 2 2 ...
 $ cls_did_eval: int [1:463] 24 86 76 77 17 35 39 55 111 40 ...
 $ cls_students: int [1:463] 43 125 125 123 20 40 44 55 195 46 ...
 $ cls_level   : Factor w/ 2 levels "lower","upper": 2 2 2 2 2 2 2 2 2 2 ...
```



Model thinking



Let's just multiply those terms.



$$\text{score} = \beta_0 + \beta_1(\text{age}) + \beta_2(\text{bty_avg}) + \beta_3(\text{age} \times \text{bty_avg}) + \epsilon$$

```
mod <- score ~ age + bty_avg + age:bty_avg  
mod <- score ~ age * bty_avg
```

Independent			
		Categorical	Continuous
Dependent	Categorical		
	Continuous		Simple regression Multiple regression (additive / nonadditive)

Results

```
dat <- evals
fit <- lm(formula = mod, data = dat)
summary(fit)
```

```
Call:
lm(formula = mod, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.9410	-0.3517	0.1231	0.4040	1.0066

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.156077	0.367797	14.019	< 2e-16 ***
age	-0.026128	0.007342	-3.559	0.000412 ***
bty_avg	-0.187800	0.075724	-2.480	0.013494 *
age:bty_avg	0.005318	0.001580	3.366	0.000827 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5287 on 459 degrees of freedom
Multiple R-squared: 0.06096, Adjusted R-squared: 0.05482
F-statistic: 9.933 on 3 and 459 DF, p-value: 2.349e-06

$$\widehat{\text{score}} = 5.16 - 0.03(\text{age}) - 0.19(\text{bty_avg}) + 0.01(\text{age} \times \text{bty_avg})$$



So the **simple main effect** of average beauty on score is **-0.19, when age is 0**. Well done, Willempie, that's illuminating. You could've centered your predictors.



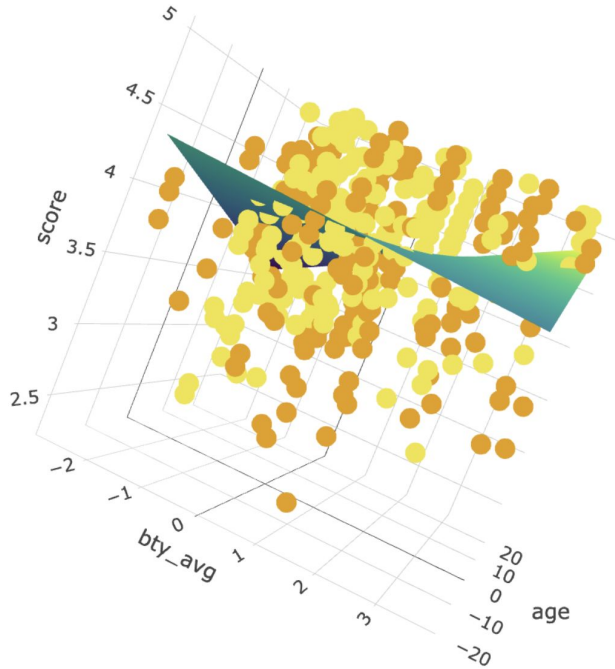
Talk to the hand, I'm only interested in the **interaction effect, anyway**.

```
dat$age <- dat$age - mean(dat$age) # 48.4
```

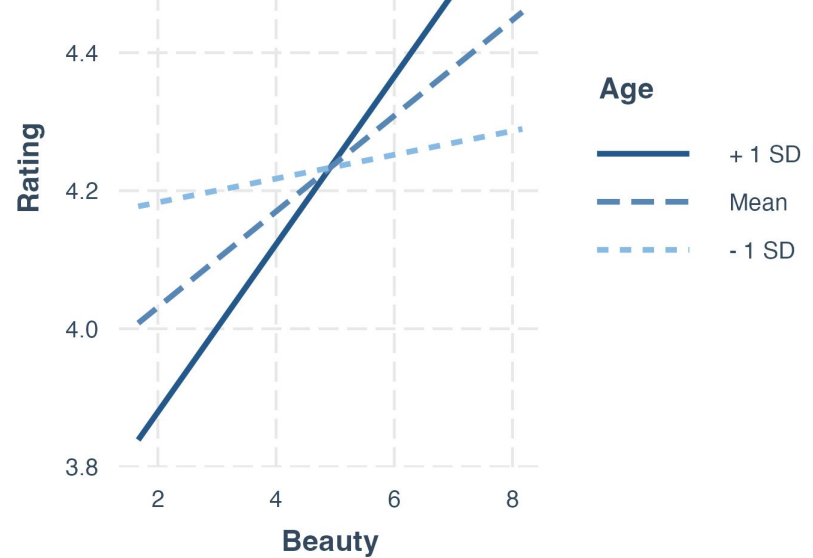


[To mean center?](#) Practical advice on last paragraph of the Discussion section.

Visualize interaction



```
library("interactions")
interactions::interact_plot(model = fit, pred =
bty_avg, modx = age, data = dat)
```



Simple slopes analysis & Johnson–Neyman interval

```
library("sandwich")
interactions::sim_slopes(fit, pred = bty_avg, modx = age)
interactions::johnson_neyman(fit, pred = bty_avg,
                              modx = age, alpha = .05)
```

JOHNSON-NEYMAN INTERVAL

When age is OUTSIDE the interval [16.72, 42.12], the slope of bty_avg is $p < .05$.

Note: The range of observed values of age is [29.00, 73.00]

SIMPLE SLOPES ANALYSIS

Slope of bty_avg when age = 38.56227 (- 1 SD):

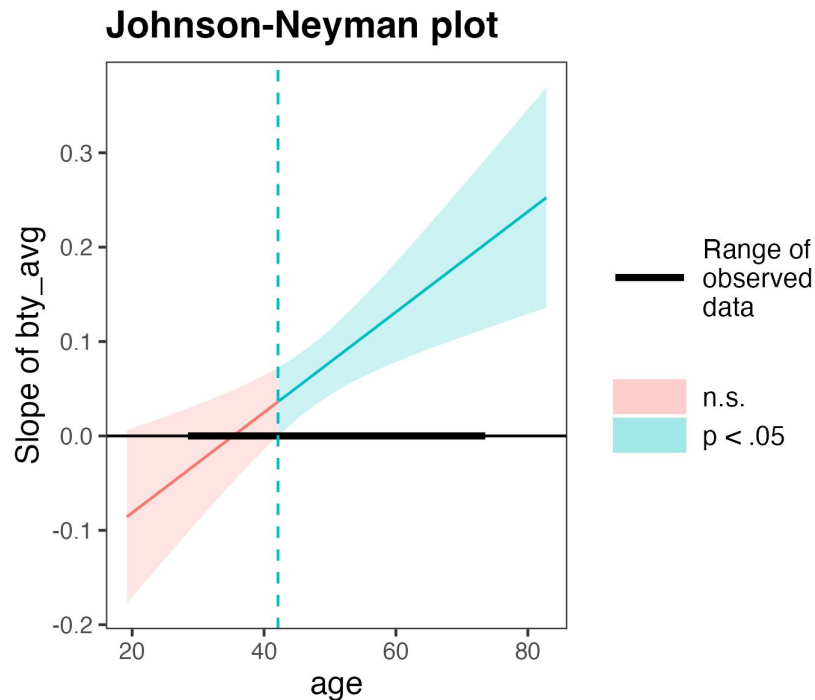
Est.	S.E.	t val.	p
0.02	0.02	0.81	0.42

Slope of bty_avg when age = 48.36501 (Mean):

Est.	S.E.	t val.	p
0.07	0.02	4.06	0.00

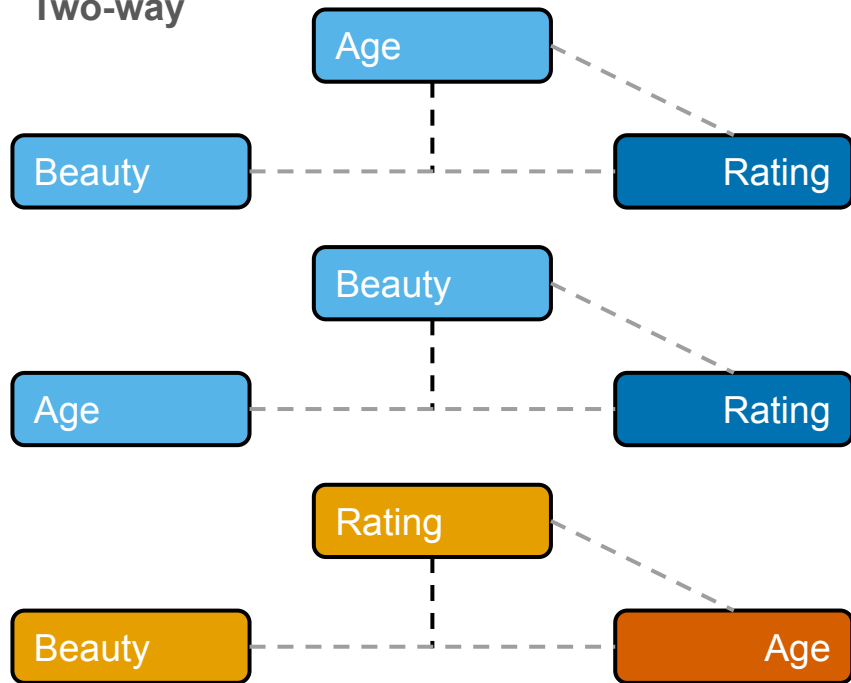
Slope of bty_avg when age = 58.16775 (+ 1 SD):

Est.	S.E.	t val.	p
0.12	0.02	4.91	0.00

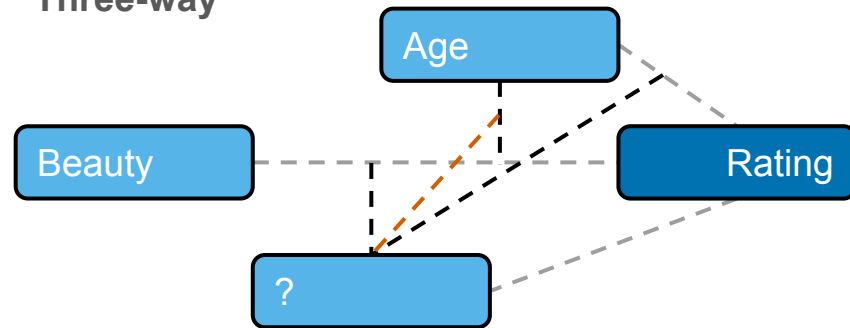


Higher-order interactions

Two-way



Three-way



`score ~ bty_avg * age * ?`

Causality



Students don't know what's best for their own learning (The Conversation)

“ Instructors who are viewed as better looking receive higher instructional ratings, [...]. This impact exists within university departments and even within particular courses, and is larger for male than for female instructors.

Disentangling whether this outcome represents productivity or discrimination is, as with the issue generally, probably impossible.

— [Hamermesh & Parker \(2005\)](#) ; [NBER](#)

— Photo by [Andrea Piacquadio](#)

Model formulae in R

```
y ~ x    # with intercept
y ~ 1 + x # with intercept
y ~ 0 + x # without intercept

y ~ x + z # add a term
y ~ x - z # remove a term
y ~ I(x + z) # sum two terms
y ~ x : z   # create an interaction term
y ~ x * z   # create crossed terms (x + z + x:z)
y ~ x %in% z) # create nested terms (x + x:z)
```

and there's more...

Traditional name	Model formula	R code
Bivariate regression	$Y \sim X1$ (continuous)	<code>lm(Y ~ X)</code>
One-way ANOVA	$Y \sim X1$ (categorical)	<code>lm(Y ~ X)</code>
Two-way ANOVA	$Y \sim X1$ (cat) + $X2$ (cat)	<code>lm(Y ~ X1 + X2)</code>
ANCOVA	$Y \sim X1$ (cat) + $X2$ (cont)	<code>lm(Y ~ X1 + X2)</code>
Multiple regression	$Y \sim X1$ (cont) + $X2$ (cont)	<code>lm(Y ~ X1 + X2)</code>
Factorial ANOVA	$Y \sim X1$ (cat) * $X2$ (cat)	<code>lm(Y ~ X1 * X2)</code> or <code>lm(Y ~ X1 + X2 + X1:X2)</code>

Table from [An Introduction to R](#)

💡 Nearly anything can be described with a [\(generalized linear\) regression model](#). A [cheat sheet](#) for model formulae. Understand the [t-test](#) and [ANOVA](#) as a linear model ([cheat sheet](#)).

Cooling Down

Untitled by Koen Derks (aRtsy package) 🎨

Takeaways

There is no such thing as too much power*

* always determine the amount of participants you need to detect a particular effect size *before* the experiment

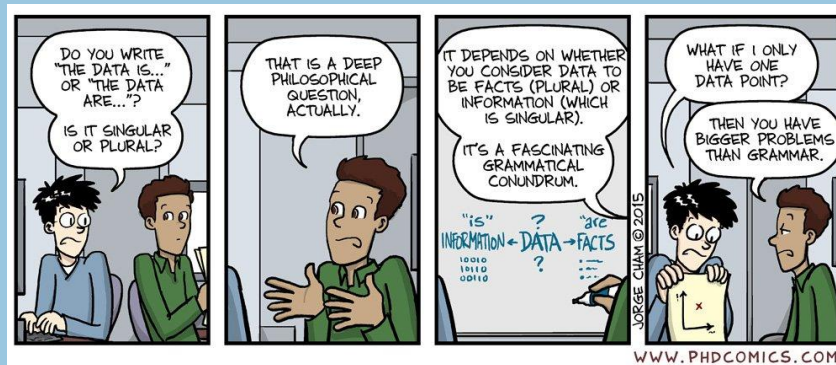


Illustration by [Jorge Cham](http://www.phdcomics.com)



Takeaways

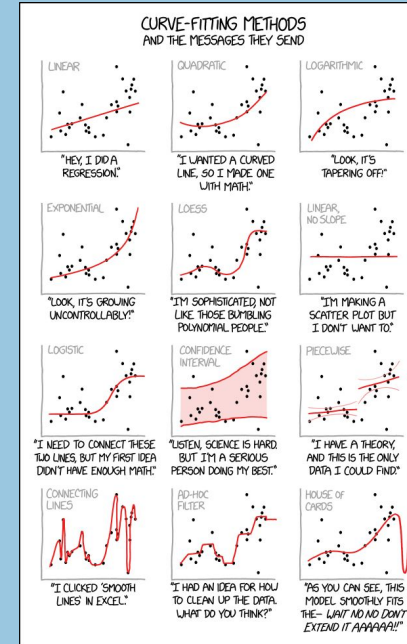


Illustration by [Randall Munroe](#) ([wtf](#))



Takeaways

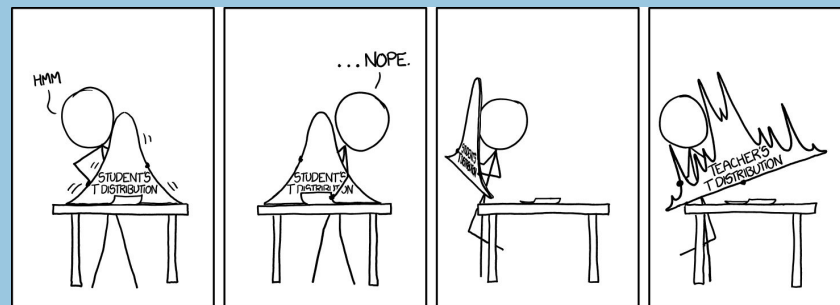


Illustration by [Randall Munroe](#) ([wtf](#))



Takeaways

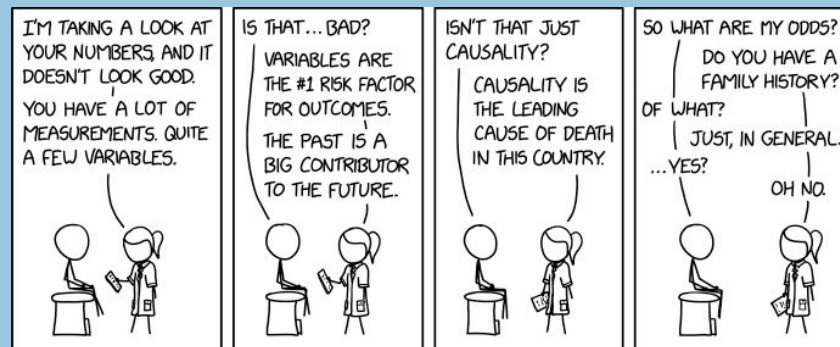


Illustration by [Randall Munroe](#) ([wtf](#))



Exam(ple) question

Je wilt een auto gebruiken, maar maakt je ook zorgen om het milieu. Je gebruikt de ``mtcars`` dataset in R om er achter te komen wat de eigenschappen van een zuinige auto zijn. Je onderzoekt hoe de relatie tussen het verbruik (``mpg``) en de paardekracht (``hp``) wordt gemodereerd door het gewicht (``wt``).

- A. Rond af op twee decimalen en rapporteer de beta-coëfficiënt van de significante interactie.
- B. Rond af op twee decimalen en rapporteer tot welk gewicht (``wt``) de paardekracht (``hp``) een negatieve relatie heeft met het verbruik (``mpg``).



This R data set is frequently used in tutorials, help files, and question-and-answer websites like [Stack Overflow](#) and the [Posit Forum](#).



Next week

Topics

Statistical reasoning with GLM

Multiple linear regression

| Dummy-variable regression

Logistic regression

Multilevel and longitudinal analysis

Bayesian statistics



Illustration by [Jennifer Cheuk](#)



Colophon

Slides

alexandersavi.nl/teaching/

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Approved for [Free Cultural Works](#).



Don't look here!

Show that a t -test and linear regression analysis return the same results.

Share your attempt (and tell whether you needed hints)!

Hints (select and copy/paste the invisible text below to reveal it)

0.

1.

2.

3.